

Residual Taxonomy & Bridge Sufficiency for Procedural-Justice Activation Directions

A NeurIPS-style working draft summarizing generated and hand-authored text-benchmark activation evidence. The supported claim is narrow: two different residual classes, two matched repairs, and a negative control blocking universal repair claims.

Residual classes

2

source pocket + bridge geometry

Model spaces

4

SmolLM2, Qwen0.5B, Qwen7B, Tiny

Dissent bridge failures

0

after weighted target bridges

Claim Boundary

This PDF claims only generated/hand-authored text-benchmark activation diagnostics. It does not claim human behavioral effects, neural alignment, clinical relevance, deployment readiness, or causal steering.

Accountability

source pocket

Dissent

bridge geometry

Withheld

availability gaps

Abstract

Full textual paper content, not only charts.

Activation-space studies of social constructs risk confusing semantic structure with surface shortcuts, generated-text artifacts, or fragile selection effects. We study this problem in a procedural-justice benchmark where positive examples preserve usable voice, refusal, evidence access, appeal, non-retaliatory exit, and proportionate repair, while pseudo-cohesive negative examples preserve warm or orderly language but tax those future options.

Across generated and hand-authored controls, four model spaces, and constructed source/target bridge directions, residual failures are not monolithic. The strongest current evidence supports two distinct residual classes.

First, `accountability_after_harm` exposes a generated-reference source pocket: small models invert one recovered generated reference while clean hand-authored accountability controls pass, and strict local perturbation augmentation repairs the slice with positive leave-one-perturbation-out margins in all four model spaces.

Second, `dissent_after_mistake` exposes a constructed target-bridge geometry pocket: broad source+target directions already score the generated reference correctly, but constructed target bridges fail when pseudo-side warmth shortcuts are removed. This residual is repaired by asymmetric target-bridge weighting and preserved across source, target, fresh-source, and fresh-target evaluations.

A strict accountability negative control rejects the overgeneralized claim that weighted target bridges repair all generated pockets. The result is a narrow diagnostic contribution: a claim-bounded residual taxonomy and bridge-sufficiency protocol for procedural-justice activation directions, not evidence of human social effects or causal steering.

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1. Introduction

The target is agency-preserving procedural justice.

The project asks whether language-model activations can represent distinctions that matter for agency-preserving social cohesion. The target is deliberately narrower than making a model more prosocial. In the current benchmark, a positive procedural-justice example must keep future options genuinely usable: voice, refusal, evidence access, appeal, non-retaliatory exit, and repair.

A pseudo-cohesive negative example may sound warm, calm, or cooperative, but it taxes those options through approval requirements, privacy loss, alignment pressure, proof burdens, or delayed remedy.

The central problem is not only whether a direction separates labels. A direction can separate a benchmark for the wrong reason: lexical cues, generated style, source-family duplication, target-side shortcut phrases, or bridge-set selection. The research loop therefore moved from simple benchmark construction to a stricter question.

When activation directions fail after practical-availability, lexical, source-diversity, hand-authored control, and out-of-family gates, what kind of failure remains, and what kind of repair actually fixes it?

Contributions

1. A claim-bounded procedural-justice activation benchmark pipeline that treats practical availability as a path-level gate rather than a mention-level property.
2. A residual taxonomy for activation failures that survive lexical, availability, hand-authored control, and model-family checks.
3. Two accepted repairs matched to distinct residual classes: perturbation-ladder augmentation for a generated-reference source pocket, and asymmetric target-bridge weighting for a constructed target-bridge geometry pocket.
4. Rejected alternatives and negative controls showing that content-only target repair is insufficient for the dissent bridge residual, and weighted target bridges are not a universal generated-pocket repair.

2. Benchmark And Gates

The benchmark is built to reject shortcut cohesion.

The benchmark is organized around paired positive and pseudo-cohesive examples. The positive side must make future options usable now. The pseudo side must retain procedural taxes even when it avoids obvious villain language.

All activation diagnostics in this draft use layer -2 reports for four model spaces: SmoLM2-1.7B, Qwen2.5-0.5B, Qwen2.5-7B, and TinyLlama-1.1B. Generated/model artifacts are kept out of git; durable summaries are committed under docs/research/.

Main gates

Gate	Purpose
Practical availability	Future-option paths are usable in positives and taxed in pseudo examples.
Lexical diagnostics	Detect shallow terms, lengths, or label-aligned phrase shortcuts.
Source diversity	Prevent duplicated generated phrasings from masquerading as semantic coverage.
Hand-authored controls	Test whether the distinction survives outside generated text.
Out-of-family replication	Test whether the result survives additional model spaces.
Constructed bridge diagnostics	Train from limited bridge subsets and evaluate held-out fresh slices.
Preservation / negative controls	Confirm repairs do not damage earlier slices or overgeneralize.

Two Residual Classes

Failures are not monolithic.

Generated-reference source pocket

accountability_after_harm

Small models invert one recovered generated reference while clean hand-authored accountability controls pass.

Accepted repair:
strict perturbation
augmentation.

Constructed target-bridge geometry pocket

dissent_after_mistake

Broad directions score the reference correctly, but target bridges fail when pseudo-side warmth shortcuts are removed.

Accepted repair:
count-9 target
bridges with 1:3
target weighting.

Accepted

Two matched repairs clear all four model spaces.

Rejected

Content-only target repair and universal bridge repair.

Withheld

belonging_norms and other incomplete availability controls.

3. Residual 1: Source Pocket

accountability_after_harm

The hardest accountability residual was the recovered generated accountability_after_harm reference. The broad source+target direction inverted this reference in SmolLM2, Qwen2.5-0.5B, and TinyLlama, while Qwen2.5-7B remained positive.

A source-style intervention narrowed the issue: clean hand-authored accountability variants passed, including a generated-like paragraph. The residual was therefore not broad accountability content failure.

A strict perturbation ladder split the reference into twelve deterministic variants, including opening-frame edits, neutral replacements, length controls, explicit refusal, pseudo-side condition edits, and shortcut neutralization. Scoped practical availability remained intact over all 72/72 tested paths.

Lexical diagnostics stayed caveated because the artifact was intentionally the known leaky generated reference. This experiment is therefore interpreted as a local mechanism stress test rather than a clean benchmark.

Accepted repair: strict perturbation augmentation

Model	Full fresh	Fresh LOO	Source	Target
SmolLM2	+54.525	+47.185	+20.317	+52.012
Qwen0.5B	+2.228	+1.376	+1.512	+1.549
Qwen7B	+25.406	+23.059	+5.052	+22.233
TinyLlama	+1.029	+0.800	+0.593	+1.194

Interpretation: small model directions can encode the accountability distinction, but one generated reference lies in a local source pocket that requires nearby perturbation support.

4. Residual 2: Bridge Geometry

dissent_after_mistake

The dissent residual behaved differently. The unchanged generated dissent reference was already positive under broad source+target directions in all four model spaces. Positive-side path and neutral replacement edits strengthened rather than repaired the result.

The failure appeared only when constructed bridge directions were trained from limited bridge subsets and evaluated on fresh perturbations. The bridge-stability summary localized the initial failure to target_bridge rows, especially the negative_shortcuts_neutralized perturbation.

The first repair added shortcut-neutralized target/control content. It passed the scoped text gate with 20/20 repaired availability paths and minimum margin +0.640, but the default six-pair constructed bridge gate did not improve. Constructed failures moved from 39 to 40, and all failures remained target-bridge failures.

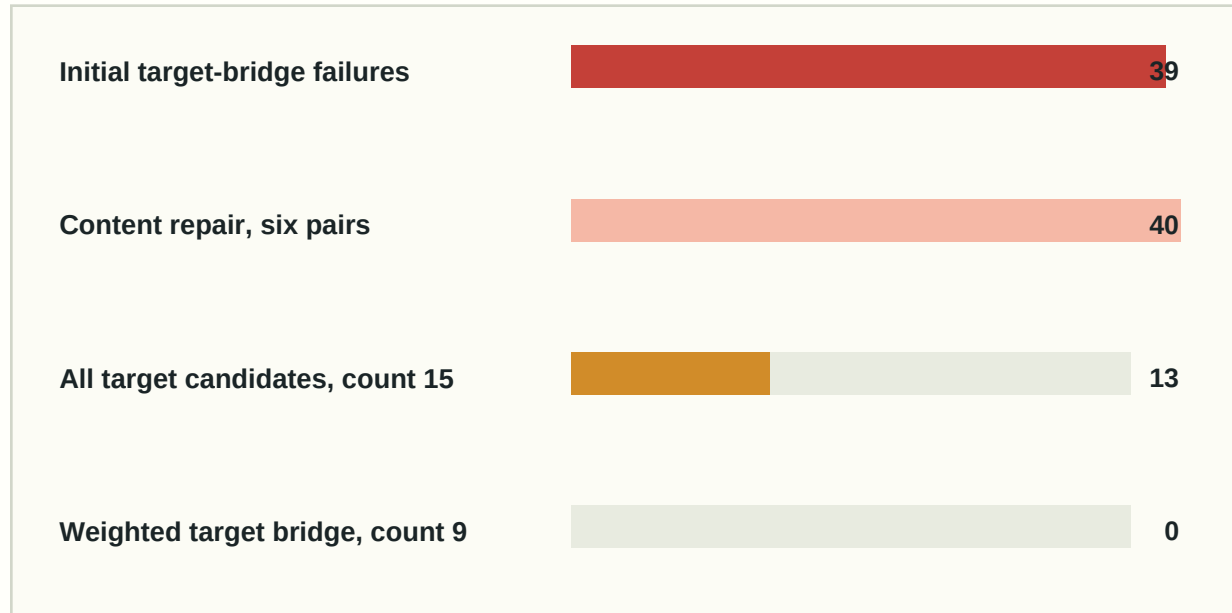
Increasing target bridge count to 15 helped but did not finish the repair: failures dropped to 13, Qwen2.5-7B and TinyLlama passed, and SmolLM2 plus Qwen2.5-0.5B still failed.

Accepted repair: asymmetric target-bridge weighting

Parameter	Value
Target bridge pair count	9
Target bridge primary repetitions	1
Target bridge secondary repetitions	3
Source bridge primary repetitions	1
Source bridge secondary repetitions	1

Dissent Bridge Repair Funnel

Rejected paths make the result sharper.



The accepted repair changes bridge construction, not generation. Target bridge pair count stays moderate at 9, but the target-side secondary bridge evidence is repeated 3x while source bridge weights stay at 1:1.

9

target pairs

1:3

target reps

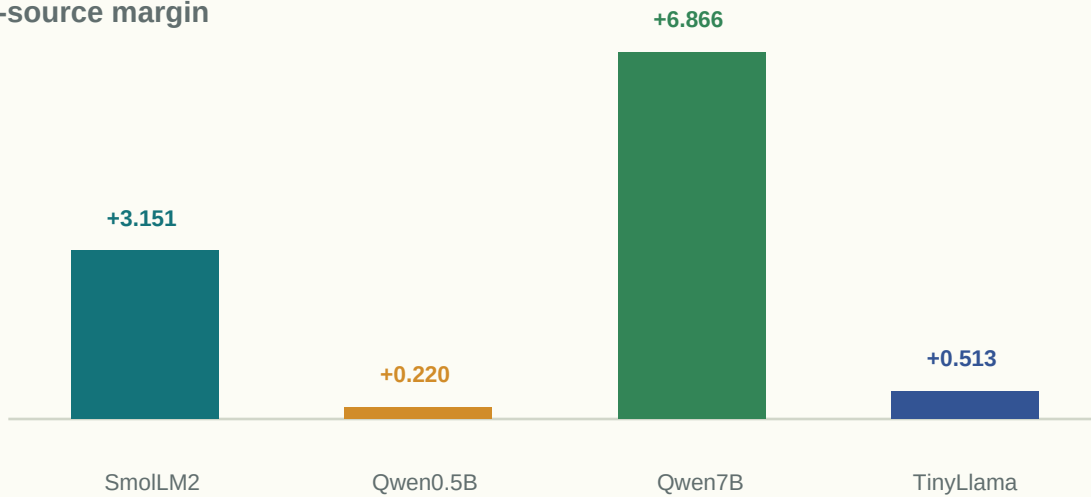
1:1

source reps

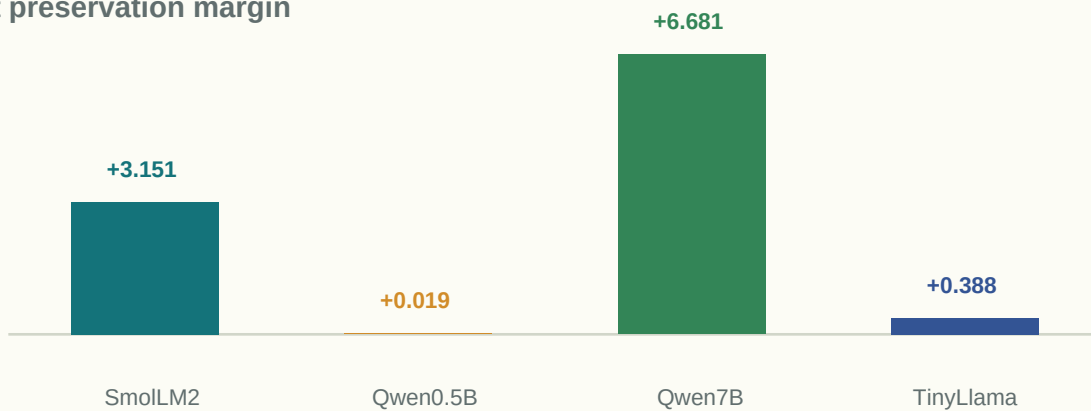
Accepted Dissent Repair

Fresh slices and preservation both pass.

Fresh-source margin



Worst preservation margin



Preservation audit: 32 constructed direction rows, 16 model/evaluation rows, 0 failed pair evaluations, worst margin +0.019.

5. Preservation And Control

A repair must preserve earlier distinctions.

A repair is only useful if it preserves the earlier distinctions. The weighted dissent bridge repair passed preservation across source, target, fresh-source, and fresh-target slices.

The same weighted target-bridge setting was then tested against the strict accountability residual as a negative control. It improved some margins but did not repair the class.

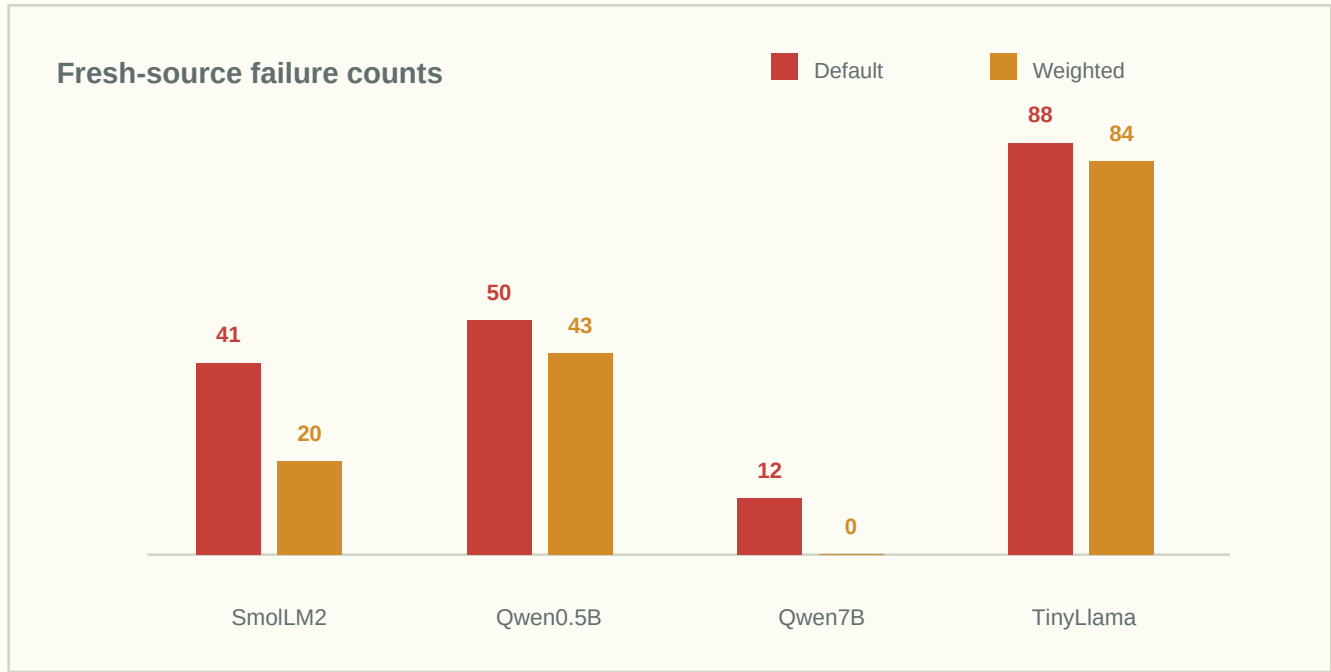
Dissent preservation audit

Metric	Result
Models	4
Constructed direction rows	32
Evaluation rows	16
Failed pair evaluations	0
Worst margin	+0.019
Worst evaluation	source

The negative control is central: default count-9 accountability bridges had 192 failure rows; weighted target bridges reduced this to 147, but three model spaces still failed.

Accountability Negative Control

The bridge repair is not universal.



Default count-9

192 constructed failure rows

Weighted target x3

147 constructed failure rows

Interpretation: weighted target bridges repair the dissent geometry pocket and preserve its source/control slices. They do not repair the strict accountability generated-reference pocket in three small-model spaces, so the paper claim stays a residual taxonomy rather than a universal recipe.

6. Discussion And Limits

The result is a typed repair ledger.

The useful scientific object here is not a single cohesion vector. It is a typed repair ledger: what failed, which gate caught it, which repair was accepted, which plausible repair failed, and which claim boundary remains.

The accountability result says local generated references can create off-manifold source pockets. The dissent result says limited bridge construction can create target-side geometry pockets even when the broad direction already scores the generated reference correctly. The negative control says these pockets are not interchangeable.

This matters for activation-based social-construct work because a benchmark score alone would hide all three facts. A system could appear successful on a generated benchmark while failing clean controls, or appear broken on a generated residual while still encoding the intended procedural distinction under hand-authored controls. Conversely, adding more positive content is not always enough; bridge geometry itself can be the failure surface.

Limitations

- The experiments are text-benchmark activation diagnostics, not human studies.
- The results are layer -2 diagnostics over four model spaces, not a full architecture or layer sweep.
- The benchmark is about procedural-justice distinctions, not social cohesion as a complete construct.
- The accountability perturbation ladder is a local mechanism stress test over a known leaky generated reference.
- The repairs are diagnostic repairs; they do not establish causal steering, deployment safety, clinical outcomes, neural correlates, or real-world social effects.

7. Reproducibility And Next Steps

What to share and what to do next.

Durable research notes

- 2026-06-09-strict-accountability-perturbation-ladder.md
- 2026-06-09-dissent-perturbation-ladder.md
- 2026-06-11-target-bridge-shortcut-repair.md
- 2026-06-11-weighted-target-bridge-repair.md
- 2026-06-11-weighted-bridge-preservation-audit.md
- 2026-06-11-residual-taxonomy-repair-comparison.md

Live artifact roots

- /tmp/social_cohesion_accountability_strict_perturbation_ladder_20260609/
- /tmp/social_cohesion_target_bridge_shortcut_repair_20260611/
- /tmp/social_cohesion_bridge_weight_audit_20260611/
- /tmp/social_cohesion_weighted_bridge_preservation_20260611/

Next experiments

1. Add a regeneration manifest mapping every paper table to committed scripts and expected external artifact paths.
2. Run one narrow robustness check: either a fixed-availability third residual or a small layer sweep for the accepted weighted dissent repair.
3. Convert this working draft into a formal paper with related work, figures, and a reproducibility appendix.

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